2) SPLITTING TRAINING DATA INTO 80% FOR TRAINING AND 20% FOR VALIDATION

```
!pip install tensorflow keras matplotlib numpy
```

```
Requirement already satisfied: tensorflow in /usr/local/lib/python3.11/dist-packages (2.18.0)
Requirement already satisfied: keras in /usr/local/lib/python3.11/dist-packages (3.8.0)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.11/dist-packages (3.10.0)
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Requirement already satisfied: absl-py>=1.0.0 in /usr/local/lib/python3.11/dist-packages (from
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/usr/local/lib/python3.11/dist-packages (from tensorflow) (1.6.3) Requirement already satisfied:
flatbuffers>=24.3.25 in /usr/local/lib/python3.11/dist-packages (from tensorflow) (25.2.10) Requirement
already satisfied: gast!=0.5.0,!=0.5.1,!=0.5.2,>=0.2.1 in /usr/local/lib/python3.11/dist-packages (from
tensorflow) (0.6 Requirement already satisfied: google-pasta>=0.1.1 in
/usr/local/lib/python3.11/dist-packages (from tensorflow) (0.2.0) Requirement already satisfied:
libclang>=13.0.0 in /usr/local/lib/python3.11/dist-packages (from tensorflow) (18.1.1) Requirement
already satisfied: opt-einsum>=2.3.2 in /usr/local/lib/python3.11/dist-packages (from tensorflow) (3.4.0)
Requirement already satisfied: packaging in /usr/local/lib/python3.11/dist-packages (from tensorflow)
(24.2) Requirement already satisfied:
protobuf!=4.21.0,!=4.21.1,!=4.21.2,!=4.21.3,!=4.21.4,!=4.21.5,<6.0.0dev,>=3.20.3 in /usr/local/lib/py
Requirement already satisfied: requests<3,>=2.21.0 in /usr/local/lib/python3.11/dist-packages (from
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/usr/local/lib/python3.11/dist-packages (from tensorflow) (1.17.0) Requirement already satisfied:
termcolor>=1.1.0 in /usr/local/lib/python3.11/dist-packages (from tensorflow) (3.0.0) Requirement
already satisfied: typing-extensions>=3.6.6 in /usr/local/lib/python3.11/dist-packages (from tensorflow)
(4.13.0) Requirement already satisfied: wrapt>=1.11.0 in /usr/local/lib/python3.11/dist-packages (from
tensorflow) (1.17.2) Requirement already satisfied: grpcio<2.0,>=1.24.3 in
/usr/local/lib/python3.11/dist-packages (from tensorflow) (1.71.0) Requirement already satisfied:
tensorboard<2.19,>=2.18 in /usr/local/lib/python3.11/dist-packages (from tensorflow) (2.18.0)
Requirement already satisfied: h5py>=3.11.0 in /usr/local/lib/python3.11/dist-packages (from tensorflow)
(3.13.0) Requirement already satisfied: ml-dtypes<0.5.0,>=0.4.0 in /usr/local/lib/python3.11/dist-packages
(from tensorflow) (0.4.1) Requirement already satisfied: tensorflow-io-gcs-filesystem>=0.23.1 in
/usr/local/lib/python3.11/dist-packages (from tensorflow) (0 Requirement already satisfied: rich in
/usr/local/lib/python3.11/dist-packages (from keras) (13.9.4)
Requirement already satisfied: namex in /usr/local/lib/python3.11/dist-packages (from keras) (0.0.8)
Requirement already satisfied: optree in /usr/local/lib/python3.11/dist-packages (from keras) (0.14.1)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.11/dist-packages (from
matplotlib) (1.3.1) Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.11/dist-packages
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/usr/local/lib/python3.11/dist-packages (from matplotlib) (4.56.0) Requirement already satisfied:
kiwisolver>=1.3.1 in /usr/local/lib/python3.11/dist-packages (from matplotlib) (1.4.8) Requirement
already satisfied: pillow>=8 in /usr/local/lib/python3.11/dist-packages (from matplotlib) (11.1.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.11/dist-packages (from
matplotlib) (3.2.3) Requirement already satisfied: python-dateutil>=2.7 in
/usr/local/lib/python3.11/dist-packages (from matplotlib) (2.8.2) Requirement already satisfied:
wheel<1.0,>=0.23.0 in /usr/local/lib/python3.11/dist-packages (from astunparse>=1.6.0->tensorflow) (0
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.11/dist-packages (from
requests<3,>=2.21.0->tensor Requirement already satisfied: idna<4,>=2.5 in
/usr/local/lib/python3.11/dist-packages (from requests<3,>=2.21.0->tensorflow) (3.10 Requirement
already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.11/dist-packages (from
```

requests<3,>=2.21.0->tensorflow) Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.11/dist-packages (from requests<3,>=2.21.0->tensorflow) Requirement already satisfied: markdown>=2.6.8 in /usr/local/lib/python3.11/dist-packages (from tensorboard<2.19,>=2.18->tensorflow Requirement already satisfied: tensorboard-data-server<0.8.0,>=0.7.0 in /usr/local/lib/python3.11/dist-packages (from tensorboard<2 Requirement already satisfied: werkzeug>=1.0.1 in /usr/local/lib/python3.11/dist-packages (from tensorboard<2.19,>=2.18->tensorflow Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.11/dist-packages (from rich->keras) (3.0.0) Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.11/dist-packages (from rich->keras) (2.18.0) Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.11/dist-packages (from markdown-it-py>=2.2.0->rich->keras) (0.1 Requirement already satisfied: MarkupSafe>=2.1.1 in /usr/local/lib/python3.11/dist-packages (from werkzeug>=1.0.1->tensorboard<2.19,

```
import os
import tensorflow as tf
from tensorflow.keras.preprocessing.image import ImageDataGenerator
import matplotlib.pyplot as plt
```

Mount path

```
from google.colab import drive

# Mount your Google Drive
drive.mount('/content/drive', force_remount=True)
```

Mounted at /content/drive

Load data (PRE-PROCESSED)

```
prep_adenocarcinoma_train = '/content/drive/My Drive/PYTHON/ML_image_process/binary_classification_adenocarcinoma/preprocessed/prep_adeno
prep_normal_train = '/content/drive/My Drive/PYTHON/ML_image_process/binary_classification_adenocarcinoma/preprocessed/prep_normal_train'

prep_adenocarcinoma_test = '/content/drive/My Drive/PYTHON/ML_image_process/binary_classification_adenocarcinoma/preprocessed/prep_adenoc
prep_normal_test = '/content/drive/My Drive/PYTHON/ML_image_process/binary_classification_adenocarcinoma/preprocessed/prep_normal_test'
```

Visualize

1) prep_adenocarcinoma_train

```
import os
import matplotlib.pyplot as plt

prep_adenocarcinoma_train_folder = '/content/drive/My

Drive/PYTHON/ML_image_process/binary_classification_adenocarcinoma/preprocessed/pr

plt.figure(figsize=(22, 12))
```

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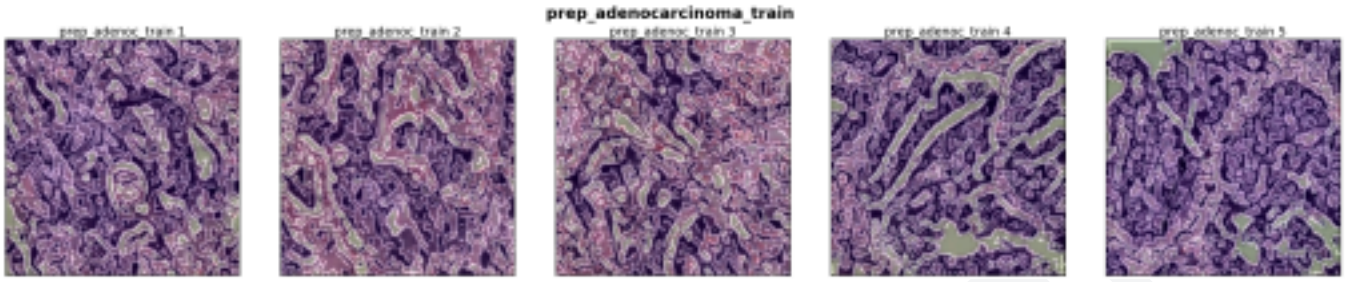
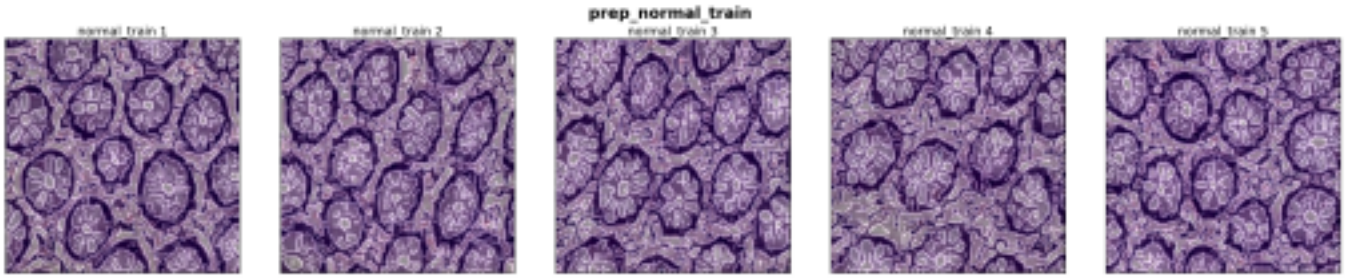
```
# Loop to display first 5 images
for i, f in enumerate(sorted(os.listdir(prep_adenocarcinoma_train_folder))[:5]):
plt.subplot(3, 5, i+1, xticks=[], yticks=[])
img = plt.imread(os.path.join(prep_adenocarcinoma_train_folder, f))
plt.imshow(img)
plt.title(f"prep_adenoc_train {i+1}", fontsize=14, pad=4, color='black')

plt.suptitle("prep_adenocarcinoma_train ", y=0.98, fontsize=18, color='black', weight='bold')
plt.tight_layout(w_pad=0.02, h_pad=0.04)
plt.show()
```



2) prep_normal_train

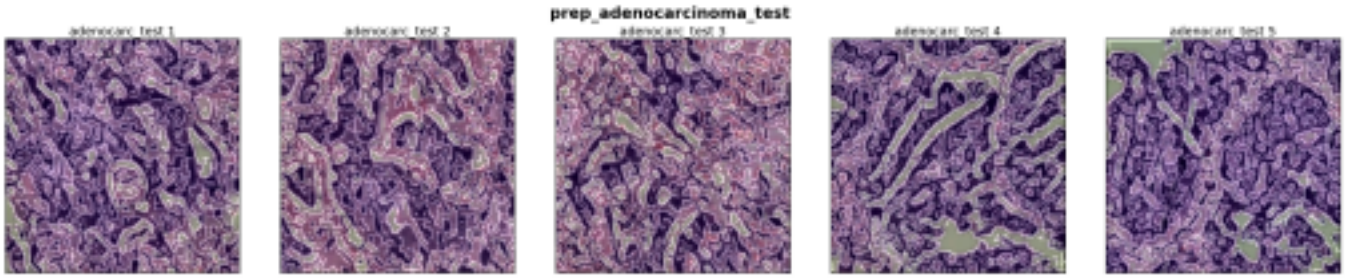
```
prep_normal_train = '/content/drive/My

Drive/PYTHON/ML_image_process/binary_classification_adenocarcinoma/preprocessed/prep_normal_train

plt.figure(figsize=(22, 12))

# Loop to display first 5 images
for i, f in enumerate(sorted(os.listdir(prep_normal_train))[:5]):
plt.subplot(3, 5, i+1, xticks=[], yticks=[])
img = plt.imread(os.path.join(prep_normal_train, f))
plt.imshow(img)
plt.title(f"normal_train {i+1}", fontsize=14, pad=4, color='black')

plt.suptitle("prep_normal_train ", y=0.98, fontsize=18, color='black', weight='bold')
plt.tight_layout(w_pad=0.02, h_pad=0.04)
plt.show()
```



3) prep_adenocarcinoma_test

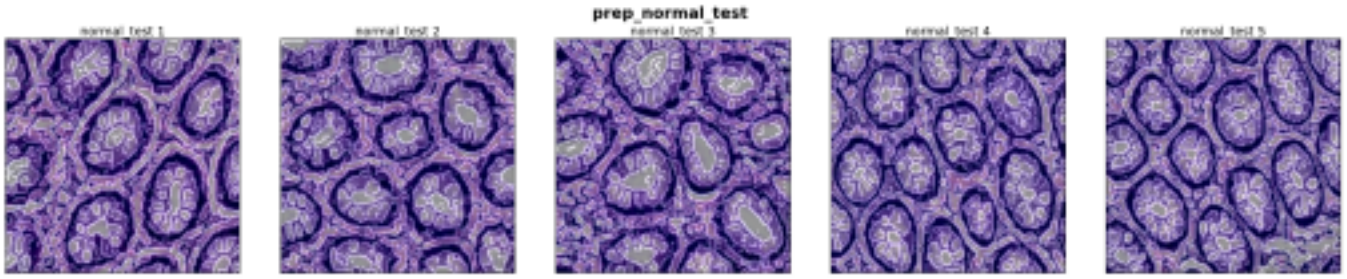
```
prep_adenocarcinoma_test = '/content/drive/My

Drive/PYTHON/ML_image_process/binary_classification_adenocarcinoma/preprocessed/prep_adeno
```

```
plt.figure(figsize=(22, 12))

# Loop to display first 5 images
for i, f in enumerate(sorted(os.listdir(prep_adenocarcinoma_test))[:5]):
    plt.subplot(3, 5, i+1, xticks=[], yticks=[])
    img = plt.imread(os.path.join(prep_adenocarcinoma_test, f))
    plt.imshow(img)
    plt.title(f"adenocarc_test {i+1}", fontsize=14, pad=4, color='black')

plt.suptitle("prep_adenocarcinoma_test ", y=0.98, fontsize=18, color='black', weight='bold')
plt.tight_layout(w_pad=0.02, h_pad=0.04)
plt.show()
```



4) prep_normal_test

```
prep_normal_test = '/content/drive/My Drive/PYTHON/ML_image_process/binary_classification_adenocarcinoma/preprocessed/prep_normal_test'
plt.figure(figsize=(22, 12))

# Loop to display first 5 images
for i, f in enumerate(sorted(os.listdir(prep_normal_test))[:5]):
    plt.subplot(3, 5, i+1, xticks=[], yticks=[])
    img = plt.imread(os.path.join(prep_normal_test, f))
    plt.imshow(img)
    plt.title(f"normal_test {i+1}", fontsize=14, pad=4, color='black')

plt.suptitle("prep_normal_test ", y=0.98, fontsize=18, color='black', weight='bold')
plt.tight_layout(w_pad=0.02, h_pad=0.04)
plt.show()
```

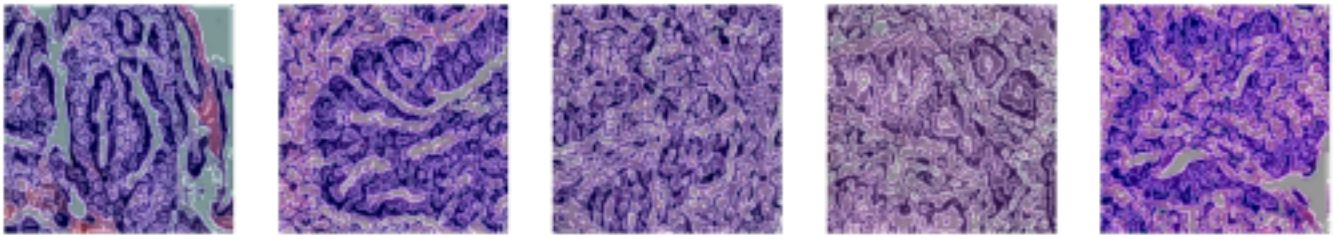


Visualize & comparison

```
import os
import cv2
import matplotlib.pyplot as plt

# Define folders
folders = {
```

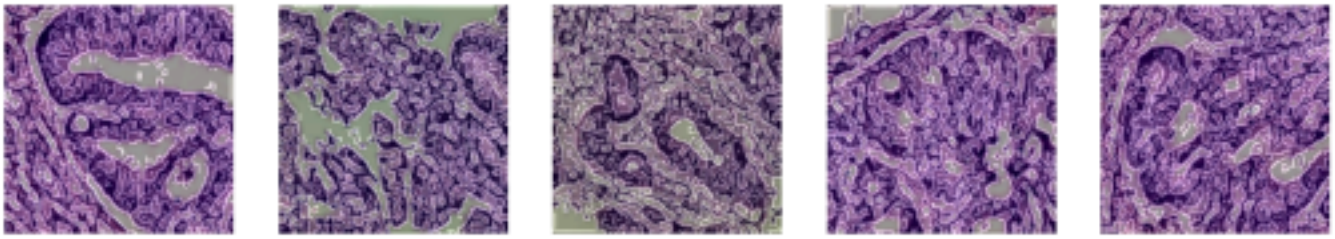
```
    "Adenocarcinoma Train": prep_adenocarcinoma_train,
    "Normal Train": prep_normal_train,
    "Adenocarcinoma Test": prep_adenocarcinoma_test,
    "Normal Test": prep_normal_test
}

# Function to display images
def show_images(folder, num=5):
    images = os.listdir(folder)[:num] # Get first 'num' images
    plt.figure(figsize=(15, 5))

    for i, img_name in enumerate(images):
        img = cv2.imread(os.path.join(folder, img_name))
        img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB) # Convert BGR to RGB

        plt.subplot(1, num, i + 1)
        plt.imshow(img)
        plt.axis("off")

    plt.show()

# Visualize images from each folder
for name, path in folders.items():
    print(f"Images: {name}")
    show_images(path)
```

Images: Adenocarcinoma Train

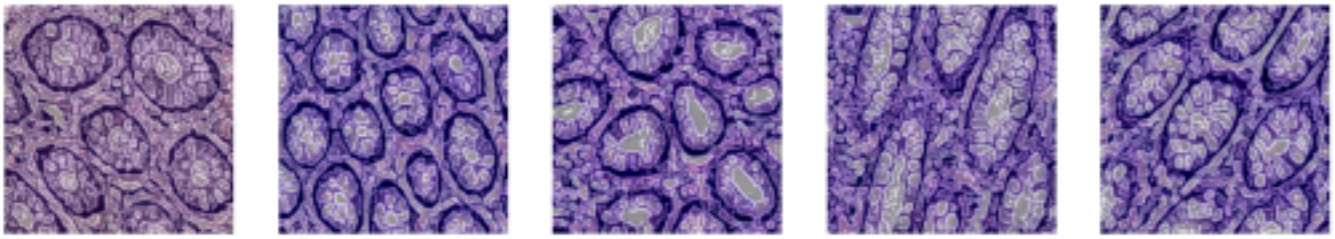


Images: Normal Train

Images: Adenocarcinoma Test



Images: Normal Test



Define paths for splitting TRAIN data into 80% train & 20% validation

```
import os
import shutil
import random

# Original train folders
prep_adenocarcinoma_train = "/content/drive/My
Drive/PYTHON/ML_image_process/binary_classification_adenocarcinoma/preprocessed/prep_aden
prep_normal_train = "/content/drive/My
Drive/PYTHON/ML_image_process/binary_classification_adenocarcinoma/preprocessed/prep_normal_train

# New folders for split data
base_dir = "/content/drive/My
Drive/PYTHON/ML_image_process/binary_classification_adenocarcinoma/split_data_TR
AIN" train_dir = os.path.join(base_dir, "80_train")
val_dir = os.path.join(base_dir, "20_val")

# Create folders
for folder in [train_dir, val_dir]:
    os.makedirs(os.path.join(folder, "adenocarcinoma"), exist_ok=True)
    os.makedirs(os.path.join(folder, "normal"), exist_ok=True)
```

Splitting

```
def split_data(source_folder, train_folder, val_folder, split_ratio=0.8):
    files = os.listdir(source_folder)
    random.shuffle(files) # Shuffle files

    split_index = int(len(files) * split_ratio)
    train_files, val_files = files[:split_index], files[split_index:]

    # Move files to respective folders
    for f in train_files:
        shutil.move(os.path.join(source_folder, f), os.path.join(train_folder, f))
    for f in val_files:
        shutil.move(os.path.join(source_folder, f), os.path.join(val_folder, f))

# Apply splitting for Adenocarcinoma and Normal classes
split_data(prep_adenocarcinoma_train, os.path.join(train_dir, "adenocarcinoma"),
os.path.join(val_dir, "adenocarcinoma")) split_data(prep_normal_train, os.path.join(train_dir,
"normal"), os.path.join(val_dir, "normal"))

print("Splitting complete! 80% data moved to train, 20% to validation.")
```

```
Splitting complete! 80% data moved to train, 20% to validation.
```

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Print the number of images in each folder:

```
import os

# Function to count images in a folder
def count_images(folder):
    return len([f for f in os.listdir(folder) if os.path.isfile(os.path.join(folder, f))])

# Define folders
folders = {
    "80_Train - Adenocarcinoma": "/content/drive/My
Drive/PYTHON/ML_image_process/binary_classification_adenocarcinoma/split_data_TRAIN/ "80_Train -
Normal": "/content/drive/My
Drive/PYTHON/ML_image_process/binary_classification_adenocarcinoma/split_data_TRAIN/80_train
"20_Validation - Adenocarcinoma": "/content/drive/My
Drive/PYTHON/ML_image_process/binary_classification_adenocarcinoma/split_data_T "20_Validation -
Normal": "/content/drive/My
Drive/PYTHON/ML_image_process/binary_classification_adenocarcinoma/split_data_TRAIN/20_
"100_Test - Adenocarcinoma": "/content/drive/My
Drive/PYTHON/ML_image_process/binary_classification_adenocarcinoma/preprocessed/prep "100_Test -
Normal": "/content/drive/My
Drive/PYTHON/ML_image_process/binary_classification_adenocarcinoma/preprocessed/prep_normal_ }

# Print image counts
for name, path in folders.items():
    print(f"{name}: {count_images(path)} images")
```

```
80_Train - Adenocarcinoma: 692 images
80_Train - Normal: 57 images
20_Validation - Adenocarcinoma: 343 images
20_Validation - Normal: 29 images
100_Test - Adenocarcinoma: 95 images
100_Test - Normal: 18 images
```

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